# Recognition of Plantation Farming Behavior Based on Deep Learning

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## Abstract

Rice and other crops' growth relies on proper farming, such as plowing and pesticide use, which are critical for yield and food safety. Plantations often manage farm activities through manual recording of farming behaviors. To address the inefficiency of manual recording, this study introduces the ED-FBNet model, based on the X3D method. It streamlines the model with pruning for edge device compatibility, enhancing performance with the ECA attention mechanism. Results show the ED-FBNet's mAP score at 93.19%, outperforming models like FWNet, Slowfast, and X3D by 3.25%, 6.37% and 8.92% respectively. Successfully deployed on-edge devices, this marks a significant step towards smart, automated agricultural production and agricultural product traceability.

## CCS Concepts

• **Computing methodologies** → Artificial intelligence; Computer vision.

## Keywords

Agricultural Management, Food Safety, Behavior Recognition, Edge Deployment

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## 1 Introduction

In China's southwest, where the terrain's complexity precludes the use of large machinery, plantations depend on manual labor and small-scale tools for agricultural production[1]. The extended growth periods of crops require essential management practices such as plowing and pesticide application to support their optimal growth. The effectiveness with which plantation workers execute these tasks has a direct impact on the agricultural output's yield and quality. Moreover, the standardization of farming operations is an integral part of modern agricultural production standards. Non-standard practices may compromise food safety and even lead to environmental issues. By systematically documenting the agricultural practices of workers and their designated areas, plantations can achieve precise and efficient crop management, standardize agricultural production, improve product traceability, and uphold stringent food safety standards.

Currently, there are two primary methods for documenting plantation workers' labor practices[2]. The first method requires workers to manually record agricultural data, a process that is time-consuming and subject to personal biases, compromising the reliability of the data. The second involves capturing video footage of farming operations and uploading this data to the cloud for subsequent behavior analysis. Nevertheless, this approach demands substantial network bandwidth and is prone to congestion issues, making it less feasible for small and medium-sized enterprises (SMEs) to document their planting operations effectively.

In recent years, video behavior recognition for identifying the work behaviors of agricultural workers has emerged. Due to the fuzziness of agricultural behavior data in plantations compared to general datasets, existing advanced behavior recognition algorithms struggle to be directly applied in the field. To address this issue, Zhao et al. integrated the CA attention mechanism into the LSTM algorithm, proposing the EPCI-LSTM model, which achieved a 94% accuracy rate on a 4-class dataset[3]. Xu et al. introduced an agricultural work behavior recognition method, FWNet, which utilizes convolutional blocks from R(2+1)D[4]and modules from residual networks to construct the network, achieving a 97% accuracy rate on the aforementioned dataset. However, these methods only consider four types of agricultural activities, making it difficult

to cover the entire range of plantation work behaviors. Moreover, transmitting video data of the fields to the cloud for classification using cloud computing resources consumes substantial network resources[2], and excessive video data may lead to processing difficulties in the cloud.

In the domain of behavior recognition datasets with more categories, Feichtenhofer et al. introduced the Slowfast series, which employs two pathways to separately learn the static and dynamic features in videos[5]. The two branches are constructed using ResNet of different depths and widths, achieving an accuracy rate of 81.8% on the more comprehensive human behavior recognition dataset, Kinetics-600[6]. Subsequently, Feichtenhofer proposed the lightweight X3D method, starting from lightweight networks like MobileNet[7] and extending the network in multiple dimensions, reaching an accuracy rate of 81.9% on the Kinetics-600 dataset[8]. Given the success of attention mechanisms in natural language processing and image recognition, Bertasius et al introduced the TimeSformer, a method for video understanding and action classification that fully utilizes self-attention mechanisms, achieving an accuracy rate of 82.2% on the Kinetics-600 dataset[9]. While convolution focuses more on local small neighborhood information, the self-attention mechanism pays more attention to long-distance target association information. Following this idea, Li et al. proposed the UniFormer architecture, which combines convolutional neural networks and attention mechanisms, obtaining an accuracy rate of 84.8% on the Kinetics-600 dataset[10].

When academic models are applied to industrial production, they often face challenges such as high latency, low throughput, and inefficiency. To address these issues, various model compression algorithms have been explored, including numerical quantization, model pruning, knowledge distillation, tensor decomposition, and the design of lightweight network structures. Currently, for behavior recognition on edge devices, Lin et al. have proposed the TSM algorithm, which obtains temporal dimension information through the integration of image information between consecutive frames, successfully deploying a gesture recognition algorithm on the NVIDIA JETSON NANO with an accuracy of 74.1% on the action classification dataset Kinetics-400[11]. Subsequently, Liu et al. introduced the Temporal Adaptive Module (TAM), which applies adaptive processing to videos to learn spatiotemporal information, achieving an accuracy of 76.1% on the Kinetics-400 dataset[12].

When applying behavior recognition algorithms to edge devices, challenges such as high computational resource consumption and insufficient accuracy are encountered. The contributions of this research are as follows: 1) The X3D model is trimmed using model pruning, reducing the number of parameters and computational load, and making the inference process less time-consuming. 2) The ECA attention mechanism is embedded into the backbone of the improved X3D, enhancing the model's accuracy. 3) Successfully applying the model to edge devices on farms plays a significant role in enhancing the yield and safety of agricultural products in plantations.

**Figure 1: Farming behavior samples.**

## 2 Materials and methods

### 2.1 Farming behavior dataset

To address the scarcity of publicly accessible video datasets on agricultural activities in farmlands, this study sourced videos of farm operations from Luliang County, Chongqing, China. To ensure applicability across diverse agricultural contexts, additional footage was collected from various video platforms, including Bilibili, Tencent Video, iQIYI, TikTok, Kuaishou, and YouTube. This effort resulted in the formation of the FBD agricultural behavior dataset, encompassing 8,308 sequences within 8 distinct categories: pesticide application, plowing, weeding, transplanting rice, fertilizing, harvesting, fruit picking, and picking. Figure 1 displays the FBD dataset.

The steps of data processing are described as follows: 1) manually editing out clips from long videos related to farming behavior; 2) extract frames from the video clip as keyframes using libraries such as OpenCV and remove frames that are not relevant to farming; 3) in a group of continuous video frames, every 10 frames with consistent action are treated as an action sequence sample, and video clips with discontinuous action are not selected as action sequence samples; 4) YOLOv5 and KNN[13] are employed to delineate and track the human body regions of the same farm operator across successive video frames; 5) remove action sequence groups artificially that have large regional shifts due to shaking during shooting; 6) a horizontal flip method is used for data enhancement of video frames.

The details of the final dataset are shown in Table 1, which involves eight categories of spaying pesticides, transplanting rice seedlings, hoeing, weeding, fertilizing, picking up potatoes, grabbing cotton, and harvesting, with 8,308 original video sequences, 7,426 used for training and the rest for testing.

### 2.2 Structure of the Network Framework

*2.2.1 Structure of the Network Framework.* The structure of the ED-FBNet proposed in this study is shown in Figure 2. Firstly, its spatial and temporal information is obtained through the Conv_BN_ReLU structure, where Conv is used to extract the information, BN (Batchnorm) layer is used to batch normalize the data to facilitate the computation, and the ReLU activation function is used to mitigate the overfitting phenomenon. Then it goes through the Feature extraction stage to learn the features, which contains 4 modules, similar to the widely used ResNet. Then after a Conv_BN_ReLU

**Table 1: Farming Behavior Dataset (FBD).**

| Class no | Class name | Original dataset | Training groups | Test groups |
|---|---|---|---|---|
| 0 | Spraying pesticides | 806 | 695 | 111 |
| 1 | Transplanting seedlings | 1078 | 982 | 96 |
| 2 | Hoeing | 1310 | 1154 | 156 |
| 3 | Weeding | 1290 | 1191 | 99 |
| 4 | Sprinkling fertilizer | 1050 | 924 | 126 |
| 5 | Picking up potatoes | 928 | 832 | 96 |
| 6 | Grabbing cotton | 918 | 837 | 81 |
| 7 | Harvesting | 928 | 811 | 117 |
| Total | | 8308 | 7426 | 882 |

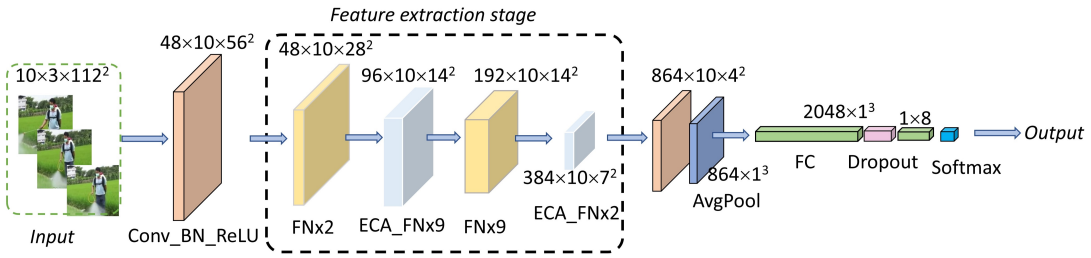


**Figure 2: The structure of ED-FBNet.**

structure, the acquired features are used for feature fusion. Then the AvgPool structure is used to downsample the overall feature information to reduce the parameter dimensions. The FC (Fully Connected) structure is then used to map the input to a dimension that is the same as the category, and the Fully Connected Layer is divided into two, and overfitting is reduced by adding the Dropout function[14] in the intermediate randomly ignoring the parameters. Eventually, a Softmax function is passed through to arrive at the final behavior classification result. The dimensions of the 4-dimensional tensor next to each block represent the size of the output feature map of that block.

In the Feature extraction stage of Figure 2, the first module is mainly used to downscale the temporal and spatial information, which consists of the two FN structures in Figure 3. The second module is used to extract the action and scene information for small localized regions and composed of nine ECA_FN structures in Figure 3. The third module is designed to extract a larger range of action and scene information and comprises nine FN structures. The last module is applied to obtain higher-level semantic information and comprises 2 ECA_FN structures.

*2.2.2 Pruning design within the framework.* The original X3D structure is shown in Figure 4. Most behavior recognition networks consist of different modules in the Feature extraction stage to learn features at multiple scales. In the behavior recognition network R(2+1)D[4], both the first and fourth feature extraction phases are composed of 2 or 3 modules. In addition to the above module composition, in the well-known classification network ResNet[15], both the first and fourth feature extraction stages consist of two or three small modules. So we pruned the first module and the fourth module in X3D and made it consist of 3 & 7 modules to both 2 or both 3 modules. The dimensions of the 4-dimensional tensor next to each block represent the size of the output feature map of that block.

During the Feature Extraction Stage, the module's receptive field expands, enhancing the semantic information acquired. In farming behavior recognition tasks, such as transplanting or hoeing, the operator's region in the image is typically fixed, with human action amplitudes being minimal and changes concentrated in small video frame regions. Consequently, the information from these small areas and their associated features are crucial. Given that R(2+1)D[4], C3D[16], and similar models maintain an equal number of modules in the middle two stages of feature extraction, this experiment sets the module ratio in the second to the third stage at 1:1. This approach ensures an equal distribution of attention-mechanism-enhanced and standard FNs across the network, facilitating a balanced integration of long-range and local features.

With the above ideas in mind, and focusing on the factors of computation and model parameter size, this study pruned the original X3D[8] and designed several structures in Table 2, with the first digit after the Model column '-' denoting the number of modules taken in the first and fourth phases, and the second and subsequent digits denoting the number of modules used in the second and third phases. For example, FB-27 indicates that the first module in the Feature Extraction Stage contains 2 FNs, the fourth module employs 2 SE_FNs, then the second employs 7 SE_FNs, and the third module employs 7 FNs.

*2.2.3 Design within the module.* Among the four modules in Figure 2 Feature extraction stage, the FN module from the X3D network

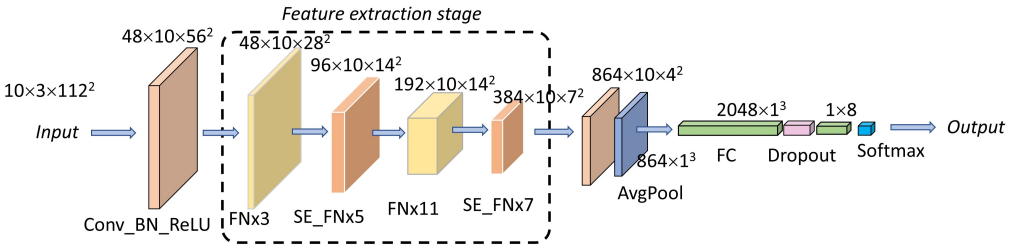


**Figure 3: The structure of X3D.**

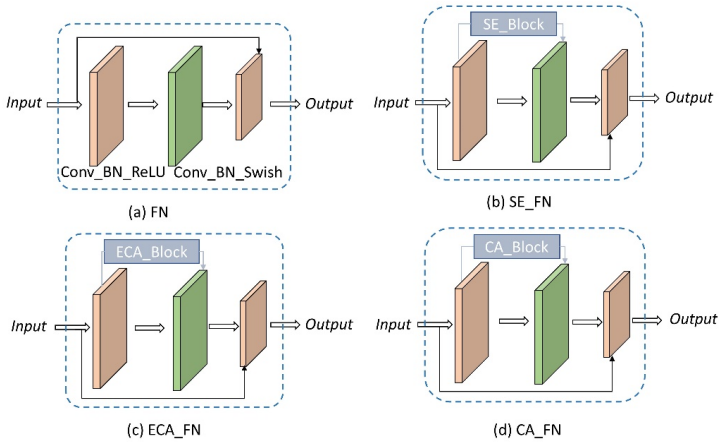


**Figure 4: The structure of FN, SE_FN, ECA_FN, CA_FN.**

**Table 2: The design of the number of Bottleneck structures in Stage.**

| Model | GFLOPs | Parameters |
|---|---|---|
| FB-27 | 2.11 | 5.22M |
| FB-37 | 2.34 | 6.02M |
| FB-28 | 2.30 | 5.44M |
| FB-29 | 2.48 | 5.71M |
| FB-38 | 2.52 | 6.24M |
| FB-39 | 2.70 | 6.51M |
| FB-210 | 2.66 | 5.93M |

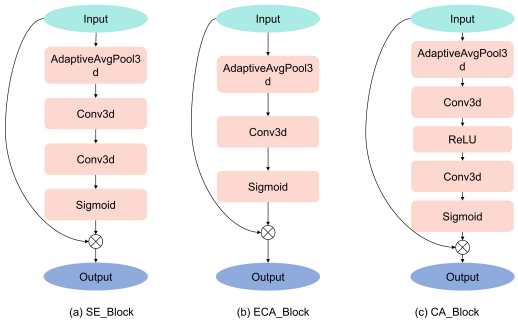


**Figure 5: The structure of SE Block, ECA Block, and CA Block.**

is used, as shown in Figure 4 (a). First through a Conv_BN_ReLU structure, then through a Conv_BN_Swish structure, the Swish activation function can be used to improve the accuracy and convergence speed of the model in some deep neural networks, and lastly, through a Conv_BN_ReLU structure with the input tensor residuals connected to the last module to alleviate the overfitting phenomenon of the model.

SE_FN is also introduced in X3D as shown in Figure 4 (b). It is an improvement of the FN structure by feeding the tensor output from the first Conv_BN_ReLU module into the SE_Block structure and then multiplying the weight tensor it outputs by the pre-Swish tensor in Conv_BN_Swish, which weights the acquired information.

Since the Resnet network with ECA obtained a more prominent performance than the one with SE attention[17], slowfast with CA attention mechanism showed better performance[18]. Under these motivations, this study constructed the ECA_ Block, CA_Block structure as shown in Figure 5(c), (d). The structure of SE_Block, ECA_Block, and CA_Block is shown in Figure 3. Meantime we tried the effect of SE_FN with FN position swapping.

**Table 3: Experimental environment.**

| | Training environment | Edge environment |
|---|---|---|
| Operating system | Windows11 | Ubuntu20 |
| CPU | i7-12700F | Quad-core ARM A57 @ 1.43 GHz |
| GPU | NVIDIA GeForce RTX 3060 | 128-core Maxwell |

**Table 4: Hyper-parameter settings.**

| Name | Value |
|---|---|
| Input size | 10×112×112×3 |
| Batch size | 12 |
| Epoch | 100 |
| Learning rate | 0.01 |
| Optimizer | SGD |

# 3 Results of experiment and discussion

## 3.1 Experimental environment

This study's experimental framework, detailed in Table 3, encompasses the model's training phase within a designated environment and subsequent evaluation of its inference capabilities on low-power edge devices. Given the imperative to balance computational capacity with cost-effectiveness, the NVIDIA JETSON NANO b01, compatible with an external CSI connection cable and camera[19], is selected as the edge device.

## 3.2 Meters design

In this study, the model parameters were refined through batch training, and the architecture was developed using the PyTorch framework. Hyperparameter settings utilized during the training process are presented in Table 4. Cross-entropy loss function was employed. Furthermore, the models underwent pre-training on the UCF101 dataset.

## 3.3 Evaluation model metrics

In terms of model evaluation, this study used mAP, Recall, and F1_score, to evaluate the performance of the model, which were calculated as shown in Equation 2), Equation 3), and Equation 4), respectively.

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

$$F1_score = \frac{2 \times Precision \times Recall}{Precison + Recall} \tag{3}$$

$$mAP = \frac{1}{n} \sum_{i=1}^{n} Precision_i \tag{4}$$

TP denotes correctly identified positive samples, FP signifies incorrectly classified negative samples as positive, TN refers to positive samples wrongly labeled as negative, and FN represents correctly detected negative samples.

## 3.4 Experimental results

*3.4.1 Exploring the different modular structures of FBNet.* In this section, various network structures designed in Table 2 are experimented with, and the results of the experiments are shown in Table 5, where the data indicate differences in performance when the structures are different. Among them, the FB-29 structure performs the highest F1-score of 92.08%, corresponding to the map is 92.48%, mRecall is 91.96%, and the amount of computation is 2.48 GFLOPs. Next is the FB-38 structure with 2.52 GFLOPs of computation, which has 87.8% F1_score, corresponding to 88.09% mAP and 87.85% Recall. While FB-27, FB-37, FB-28, FB-38, FB-39, and FB-210 are 8.88%, 6.89%, 4.63%, 4.28%, 6.75%, and 6.98% lower in mF1-score compared to ED-FBNet, respectively.

It shows the outstanding learning ability of FB-29, combining F1-score, computation GFLOPs, and other factors.

*3.4.2 Exploring the location and types of attention mechanisms.* The above FB-29 is marked as SE, then we tried different positions of the SE attention mechanism, i.e., swapping the FN of SE_FN in the Feature extraction stage, which is denoted as POS, different attention mechanisms are also used for experiments, such as ECA, CA attention mechanisms, marked as ECA, CA.

The loss values and accuracy rates during model training are shown in Figure 6. It can be found that the model with ECA_Block has lower loss values and higher accuracy during training, while the performance of the other three structures is not much different.

The experimental results of different attention mechanisms are shown in Table 6. It can be found that positional changes make the model behave worse. Among the models with other attentional mechanisms, the ECA attentional mechanism demonstrated higher recognition accuracy, and it outperformed the SE by 1.01% in the F1_score and outperformed the CA by 1.96%. Thus, the ECA was determined to be ED-FBNet.

*3.4.3 Exploring the performance of different networks in terms of accuracy.* To ascertain the reliability of ED-FBNet, this study develops a farming behavior recognition model, FWNet, alongside two established lightweight behavior recognition networks, SlowFast and X3D, for comparative analysis. The F1 scores of these four models across eight categories of farming behaviors are presented in Table 7. The results indicate that the proficiency of these models in recognizing categories 0 and 3 is comparatively lower, whereas their performance in the remaining categories achieves a success rate of 90%.

The accuracy performance of the four models was evaluated using mAP, Recall, and F1_score. The visualization results are shown in Figure 7, and it can be found that ED-FBNet achieves a F1_score of 93.09%, which is 3.25%, 6.37% and 8.92% higher than

**Table 5: Comparison of the performance of various FBNet structures.**

| Model | mAP(%) | Recall(%) | F1_score(%) | Parameters(M) | GFLOPs |
|---|---|---|---|---|---|
| FB-27 | 84.00 | 82.98 | 83.2 | 5.22 | 2.11 |
| FB-37 | 85.67 | 85.12 | 85.19 | 6.02 | 2.34 |
| FB-28 | 88.59 | 87.75 | 87.45 | 5.45 | 2.30 |
| FB-38 | 88.09 | 87.85 | 87.8 | 6.24 | 2.52 |
| FB-29 | 92.48 | 91.96 | 92.08 | 5.71 | 2.48 |
| FB-39 | 85.82 | 85.41 | 85.33 | 6.51 | 2.70 |
| FB-210 | 85.21 | 85.2 | 85.1 | 5.93 | 2.67 |

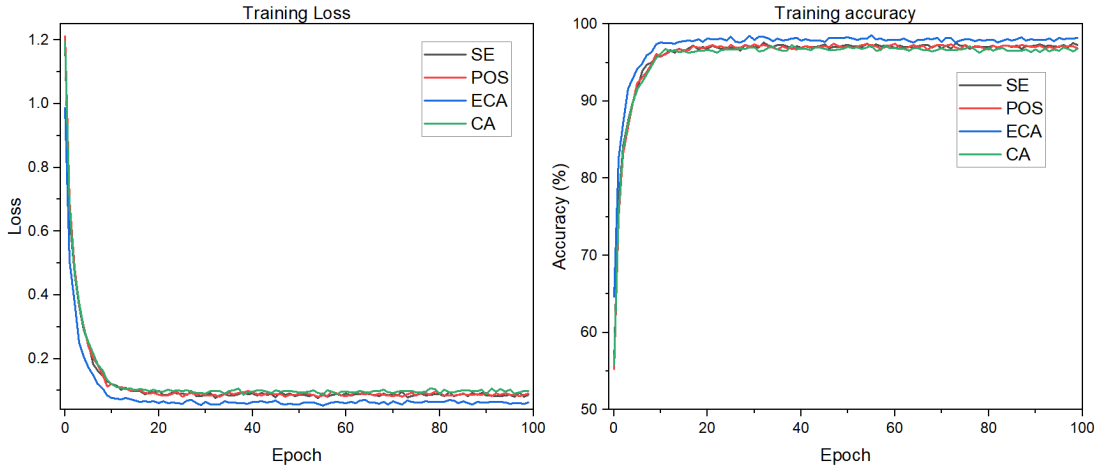


**Figure 6: The loss and accuracy during SE, POS, ECA, and CA training.**

**Table 6: Ablation experiments on attentional mechanisms.**

| | mAP(%) | Recall(%) | F1_score(%) | Params(M) | GFLOPs |
|---|---|---|---|---|---|
| SE | 92.48 | 91.96 | 92.08 | 5.71 | 2.48 |
| POS | 90.01 | 89.47 | 89.62 | 9.63 | 2.94 |
| ECA | 93.19 | 93.11 | 93.09 | 7.36 | 2.49 |
| CA | 91.41 | 90.92 | 91.13 | 5.67 | 2.46 |

**Table 7: F1_score of four farming behavior recognition methods on eight categories.**

| Method | Class no | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| SlowFast[5] | 76.82 | 88.37 | 93.7 | 72.88 | 92.21 | 84.48 | 88.07 | 76.83 |
| X3D[8] | 84.21 | 86.4 | 83.25 | 78.68 | 90.59 | 89.76 | 92.98 | 87.9 |
| FWNet[2] | 87.32 | 91.93 | 84.76 | 84.61 | 90.29 | 93.03 | 94.64 | 92.21 |
| ED-FBNet | 88.31 | 98.42 | 93.66 | 87.30 | 94.12 | 92.91 | 96.36 | 93.67 |

FWNet, X3D, and Slowfast models, respectively. And Recall as well as mAP also demonstrate the reliability of ED-FBNet.

As indicated in Table 8, the comprehensive performance of these models demonstrates that ED-FBNet achieves exceptional performance with fewer parameters and computational requirements.

*3.4.4 Exploring the performance of different networks in terms of inference speed.* The inference speed of the model is measured by the loading time and inference time. The test results are shown in Figure 8. It can be seen that in terms of model loading time, FWNet takes a shorter time than ED-FBNet, with a difference of 0.2s, and

**Table 8: Comparison of inference accuracy of different behavior recognition.**

| Method | Acc | mAP(%) | Recall(%) | F1_score(%) | Params(M) |
|---|---|---|---|---|---|
| Slowfast[5] | 84.86 | 85.17 | 83.81 | 84.17 | 61.92 |
| X3D[8] | 86.69 | 86.85 | 86.82 | 86.72 | 9.63 |
| FWNet[2] | 89.46 | 90.0 | 89.84 | 89.84 | 3.2 |
| ED-FBNet | 93.03 | 93.19 | 93.11 | 93.09 | 7.36 |

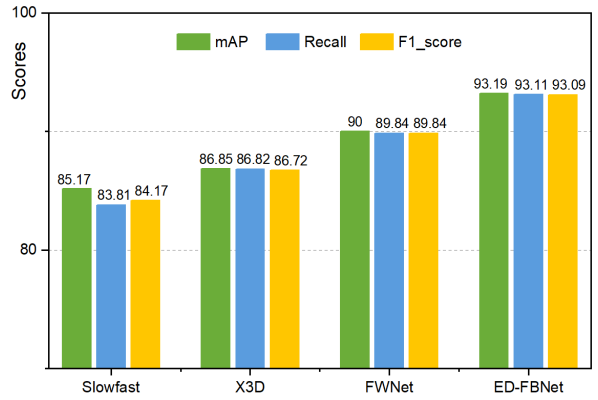


**Figure 7: Comparison of inference accuracy of different behavior recognition networks.**

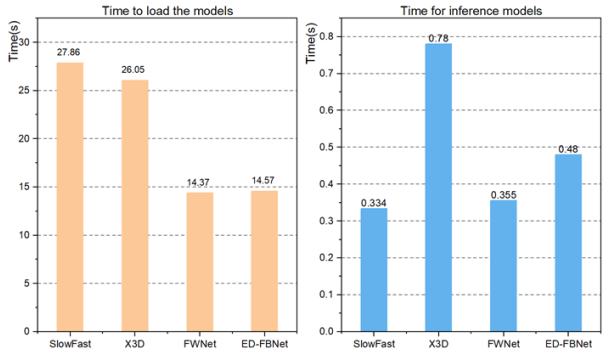


**Figure 8: The speed of SlowFast, X3D, FWNet, ED-FBNet.**

in terms of model inference time, Slowfast and FWNet take less time than ED-FBNet, with a difference of no more than 0.2s, which are both in the acceptable range.

Table 9 shows more detailed parameters of the models. The results show that the model loading time becomes longer as the number of parameters grows. While the inference time varies from model to model, the experimental results show that the correlation between the computational amount and the inference time is low. It can be seen that evaluating the models by the amount of computation and the size of the model is not completely reliable.

*3.4.5 Comprehensive performance of different models.* Combining Tables 8 and 9, the computational requirement of the X3D-based ED-FBNet is reduced by 15.3% compared to the original X3D, with an accuracy improvement of 6.34% and a reduction in inference time by 0.3s. When compared with other models, it is observed that SlowFast has the shortest inference time, while FWNet has the shortest loading time. Despite these advantages, the overall efficacy of these models falls short of that demonstrated by ED-FBNet. In summary, it can be concluded that ED-FBNet is more suitable for agricultural behavior recognition on edge devices in plantations.

## 4 Conclusion

With the advancement of precision agriculture and the development of edge computing technologies, integrating artificial intelligence into practical applications has become feasible. To efficiently record agricultural production information, thereby facilitating the traceability of agricultural products and the management and planning of farm operations, this study introduces an automated farming behavior recognition method deployed on edge devices, ED-FBNet, based on X3D, achieving the highest accuracy rate of 93.03%. Furthermore, we have also trained various models for comparison, such as SlowFast, X3D, FWNet, FB-27, FB-37, FB-28, and FB-210. This method is not only applicable to the agricultural sector but also offers valuable insights for other industries. It holds immense significance for agricultural product traceability, food safety, and the standardization and automation of production in the farming industry, as well as enhancing the comprehensiveness and efficiency of agricultural production. Future work will focus on more precise operation detection to achieve a more comprehensive system for agricultural product traceability.

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**Table 9: Performance in terms of inference speed of four farming behavior recognition networks.**

| Model | Performance | | | |
|---|---|---|---|---|
| | Params(M) | Load time(s) | Inference time(s) | GFLOPs |
| SlowFast[5] | 61.92 | 27.86 | 0.334 | 3.51 |
| X3D[8] | 9.63 | 26.05 | 0.780 | 2.94 |
| FWNet[2] | 3.2 | 14.37 | 0.355 | 1.53 |
| ED-FBNet | 7.36 | 14.57 | 0.480 | 2.49 |


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